

Garage ERP — Module & Flow Guide

Frontend status before backend and database work · 04 August 2026

How to read this. Every module carries one of three labels. **LIVE** means it works — you can enter data, it is validated, it persists, and other screens react to it. **STATIC** means the screen is built and navigable but shows sample rows and does not save. **NOT BUILT** means there is no screen at all. Nothing here is described as finished unless it has been tested.

6

Live modules

34

Static screens

34

Working routes

363

Automated checks
passing

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1. Running the app

STEP	COMMAND
Install	<code>npm install</code> from the project root
Start	<code>npm run dev</code> — opens on <code>http://localhost:5173</code>
Run all checks	<code>npm test</code> — 363 assertions
Type check	<code>npm run typecheck</code>
Production build	<code>npm run build</code>

Data lives in the browser (localStorage) until the backend exists. Clearing site data resets everything to the seeded demo set. Nothing is shared between machines yet.

2. Module map

The sidebar is the whole product. Every item below appears there, in this order.

MODULE	STATUS	WHAT IT DOES TODAY
Dashboard	LIVE	Counts of employees, customers, products and services, all computed from real records. Job calendar by expected delivery date. Setup wizard tracks genuine completion.
Inventory Supplier · Product · Purchase · Stock	LIVE	Products and suppliers with full create/edit. Stock ledger with reconciliation. Purchase screens are static.
Users Customers · Employees · Support · Accountants · Branch Admin	LIVE	All five are live with full create, edit and delete. The four staff roles share one record type rather than four near-identical tables, so a fix in one is a fix in all. Anyone named on a job card cannot be deleted — that history would lose the person who did the work.
Vehicles	LIVE / STATIC	Vehicle records are live and linked to customers. Types, brands, models and colours are static reference lists.
Services	STATIC	Service master list. Not yet read by the job card.
Quotation	LIVE	Raise a quote before any work is agreed, add labour and parts, send it, record the answer, and convert an accepted quote straight into a job card with every line copied.
Invoices	LIVE	Invoices are generated from job cards, numbered per financial year, and printable.
Job Card	LIVE	The complete workflow, start to finish. Detailed in section 3.
Accounts Tax · Payment · Expenses · Income	LIVE / STATIC	Receivables, ageing and collection are live and derived from job cards. Tax rates, expenses and income are static.
Part Sells	STATIC	Counter-sale screen. No stock movement.
Membership	STATIC	Plans, members and renewals due. Not yet applied to a job card.
Compliances	STATIC	Insurance and PUC tracking. Not surfaced at check-in.
Reports	STATIC	Service report layout only.

Email Templates	STATIC	No mail, SMS or WhatsApp gateway is connected.
Custom Fields	STATIC	Defined but not applied to any form.
Observation Library	STATIC	Reusable observations. The job card's Customer Voice list is separate and hardcoded.
Notes · Branch · Help · Settings	STATIC	Layout only.
Bay Board	NOT BUILT	Bay is a field on the job card with an occupancy guard, but there is no board view.
Inspection	NOT BUILT	Specified as a lifecycle stage. No screen, no entity, no status.
Quality Check	NOT BUILT	Specified as gating delivery. Only a <i>delivery</i> handover checklist exists.

3. Workshop — the complete module

This is the one module built end to end. A job card moves through a **state machine**: each status has a defined set of legal next statuses, and nothing can write a status directly. An action that is not allowed yet is disabled with the reason shown, rather than failing later.



Creating a job card

Sidebar → **Job Card** → **Add**, or press **C** then **J**.

SECTION	WHAT YOU ENTER
Header	Job card type, status (always opens as Draft), auto-generated number, date
Customer Details	One search box across vehicle number, mobile and name; New registers a walk-in. Then email, pickup and delivery address
Job Card Info	KM reading, fuel level, mechanic, supervisor, driver name and mobile
Five buttons	Voice what the customer reported · Parts accessories checked in · Advice recommendations · Dents existing damage · Photos condition images
Item buckets	Jobs, Spare, Lube, Outsource. Click a tile to choose where new lines land
Item search	Searches products by name, code or part number. Anything not in the catalogue can be added as a custom line
Job Card Details	Cost estimate, delivery date and time, reminder km / period / priority, work note, exit note, SMS alert
Billing	Per-bucket totals, a discount in rupees or percent, bill amount, paid, balance due. Advances can be recorded before the card is saved

Note. The whole card is assembled on screen and written in one go when you press Create. Abandoning the form leaves nothing behind.

Working the job card

Opening a job card gives six tabs. A bar at the top always names the single next step.

TAB	PURPOSE
Overview	Customer, vehicle, status, assignment, complaint summary
Estimate	Build the quote, send it, record approval or rejection
Items	Labour and parts; issue stock against a line
Invoice	Generate the invoice, take payment
Delivery	Handover checklist, gate pass, hand over
Timeline	Every event, who did it and when

4. Inventory

Stock is held as a **ledger of movements**, not a number that gets edited. Every receipt, issue and adjustment is a row. The current balance is the sum of that ledger. The figure stored on the product is only a cache.

The Inventory dashboard **reconciles** the cached figure against the ledger every time it loads, and names any product where the two disagree. If stock is ever wrong, you can see exactly which movement caused it.

Available stock

`ON HAND - RESERVED = AVAILABLE` . Parts reserved for an open job card are not offered to another.

5. Finance

Receivables are **derived**, never stored. What a customer owes is computed from invoices and receipts on demand. A balance therefore cannot drift away from the documents behind it, and there is no figure to correct by hand.

- **Receivables** — outstanding by customer, with ageing buckets
 - **Transactions** — every receipt, by mode
 - **Dashboard** — collection by payment mode, outstanding totals
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6. Customers & Vehicles

- Customers: individual or business, GSTIN, credit limit, full create and edit
 - Vehicles belong to a customer; one customer may hold many
 - Both can be added mid-flow from the job card without losing the form
 - A customer's page shows their vehicles and job card history
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7. Static modules — what to expect

Important. The 44 static screens are convincing. They have real column headings, realistic sample rows and working add forms — but **nothing saves**. They exist to agree the shape of each module before it is built. Every one of them says so on the page.

Recognising them: the sidebar tag, a banner on the page, and a footer note on the list.

8. End-to-end flows

A. The customer lifecycle

From the first phone call to a returning regular. The badge on each step says whether it works today.

- 1 Enquiry or walk-in** LIVE
Someone arrives or calls. Search by mobile, name or vehicle number to see whether they already exist — a returning customer is never entered twice.
- 2 Register the customer** LIVE
Individual or business, mobile, email, address, GSTIN and credit limit. Can be done from the job card form without losing what is already typed.
- 3 Register their vehicle** LIVE
Registration number, make, model, fuel, chassis and engine number. One customer can hold many vehicles; the job card always asks which one.
- 4 First visit — a job card** LIVE
The vehicle's journey begins. See flow B below.
- 5 They approve, pay and collect** LIVE
Estimate approval, invoice, payment and gate pass are all recorded against the customer, not just the vehicle.
- 6 History builds** LIVE
The customer page shows every vehicle they own, every job card, and what they owe. On the next visit the advisor sees what was done last time.
- 7 Service reminder is set** PARTIAL
A reminder km and period are captured on the job card and stored. Nothing sends them yet — no SMS or email gateway is connected.
- 8 They return** LIVE
Searching their mobile brings up the customer, their vehicles and their history in one step.

9 Membership and loyalty PENDING

Plans, members and renewals exist as screens, but a member's discount and free-service count are not yet applied to a job card.

10 Compliance reminders PENDING

Insurance and PUC expiry are not surfaced at check-in, which is where they would be useful.

What the system knows about a customer today

HELD AGAINST THE CUSTOMER	WHERE YOU SEE IT
Identity, contact, address, GSTIN	Customer record
Every vehicle they own	Customer page → Vehicles
Every job card, past and open	Customer page → History
What they owe, and how overdue	Finance → Receivables, with ageing
Everything paid, and by what method	Finance → Transactions
Credit limit	Customer record — recorded, not yet enforced at invoicing

B. Quotation to job card



A quotation is an offer, not a commitment: it moves no stock and touches no money. Lines lock once it is sent, so a quoted price cannot change under the customer. Converting copies every line onto a new job card and links the two, and the quotation becomes a read-only historical document.

Validity is enforced by date rather than remembered by a person — a quote past its date is flagged stale in the list and on the page, so an old price is never sent as a live offer.

C. Vehicle in, vehicle out

1 Book the vehicle in

Job Card → Add. Search the vehicle or register a new customer. Record km and fuel, tick accessories, note existing dents, take photos.

2 Complete check-in

Confirms the vehicle is physically received. The card moves out of Draft.

3 Build the estimate

Add labour and parts. Parts search shows live availability.

4 Send for approval

The card waits at Approval Pending. Items cannot be edited while the customer is deciding.

5 Record the decision

Approved, or rejected with a reason.

6 Assign and repair

Assign a technician and a bay. A bay already in use cannot be picked twice.

7 Issue parts

Stock moves out of inventory and onto the job card in one step. If stock is short, the issue fails and nothing changes.

8 Complete the repair

Extra work added during repair is marked as additional, not part of the original estimate.

9 Invoice

Numbered for the financial year. Totals come from the lines; nothing is retyped.

10 Take payment

Full or partial, by any mode. Status follows the balance automatically.

11 Deliver

Complete the handover checklist, issue the gate pass, hand over. Delivery is blocked until the checklist is complete.

D. Parts from purchase to invoice



Each arrow writes a ledger row. The stock ledger screen shows the full history for any product.

E. Money in



No step is entered twice. The receivable is a calculation, not a record.

9. Rules that govern the data

RULE	WHY IT MATTERS
Money is stored in paise	Whole numbers only. Rupee decimals accumulate rounding error over thousands of lines.
Status changes go through the machine	No screen can set a status directly, so a job card cannot reach a state it has no path to.
The ledger is the truth	Stock is the sum of its movements. The stored number is a cache, and it is checked.
Receivables are derived	Nothing to reconcile, because there is no second copy.
Issuing parts is all-or-nothing	Ledger, balance and job card line move together, or none of them do.
Every record carries company, branch and financial year	Written from day one so multi-branch is never blocked later.

10. Known gaps

Specified but not built

- **Inspection** — a lifecycle stage with no implementation
- **Quality Check** — spec'd to gate delivery; only a handover checklist exists
- **Bay Board** — bay exists as a field, not as a board

- **Rework loop** — the state machine has no backward transition. Adding QC later means changing the machine, not just adding a screen. Worth deciding before data exists.

Built but not connected

- Membership plan discount and free-service count are not applied to a job card
- Customer Voice uses a hardcoded list instead of the Observation Library
- The Services and Tax Rate masters are not read; custom lines assume 18%
- Compliance expiry is not surfaced at check-in
- Custom Fields are defined but applied nowhere

Environment limits

- No server or database — data is per-browser and lost when site data is cleared
- Photos are held in the browser; there is no file storage
- No SMS, WhatsApp or email is actually sent
- No authentication, roles or audit trail beyond the job card timeline

Not yet verified by hand

- Create Job Card compiles, serves and passes type checking, but has not been clicked through end to end with every field

11. How it is tested

`npm test` runs four suites, 363 assertions, no browser required.

SUITE	CHECKS	COVERS
Domain	47	Money arithmetic, totals, tax, state machine transitions, guards
Workshop flow	163	A whole vehicle visit, booking to delivery, with stock and ledger reconciliation
Navigation	119	Every path resolves to the right menu item and opens the right group
Quotation	34	State machine, conversion guards, expiry, and flat-vs-percentage discounts

The navigation suite was written after the sidebar highlight was reported broken. It found two real faults immediately: eleven pages were highlighting Dashboard instead of themselves, and groups stayed collapsed after navigating. Both are fixed.

